AI-Powered Airline Meal Prediction and Food ...

Submission Date

Jun 20, 2026 12:37 AM

Project Title

AI-Powered Airline Meal Prediction and Food Waste Optimization System

Brief Description of the Problem Addressed

This project focuses on developing an AI-powered decision support system for airline catering operations. The system predicts meal requirements and expected food waste percentages based on flight-related information, helping airline catering companies improve planning accuracy, reduce operational costs, and minimize food waste.

The solution was developed using Machine Learning techniques trained on a dataset containing 1,000 airline flight records. The trained model was deployed through a Hugging Face API and integrated into an intelligent conversational interface that allows users to interact with the prediction engine in a simple and efficient manner.

Data Collection Methods Used

The model was trained using historical airline flight data containing features such as:

Number of passengers
Flight duration
Aircraft type
Departure time
Flight category
Catering requirements

A total of 1,000 flight records were used for model development and evaluation.

Data Preprocessing and Cleaning Techniques Applied

Data Preprocessing and Cleaning Techniques Applied

Before training the machine learning models, the flight dataset was cleaned and preprocessed to improve data quality and prediction accuracy. The process included:

Removing duplicate records.
Handling missing values.
Encoding categorical variables using Label Encoding.
Scaling numerical features using StandardScaler.
Selecting relevant features for prediction.
Splitting the dataset into training and testing sets.

These techniques helped improve model performance and ensure reliable meal demand and food waste predictions.

Model Training Details (Algorithms, Frameworks, Parameters)

Several machine learning models were trained and compared to predict airline meal demand and food waste percentage. The training process focused on selecting the most accurate and efficient model for deployment.

Algorithms Used:

Random Forest Regressor
Gradient Boosting Regressor
XGBoost Regressor
Multi-Layer Perceptron (MLP Neural Network)

Frameworks & Libraries:

Scikit-learn for traditional ML models
XGBoost library for gradient boosting optimization
TensorFlow / Scikit-learn (MLP implementation)
Pandas & NumPy for data processing

Training Parameters (General):

Train-test split: 80/20
Cross-validation used for model evaluation
Hyperparameter tuning for optimizing performance (e.g., number of estimators, learning rate, tree depth)
Loss functions: Mean Squared Error (MSE)
Evaluation metric: R^2 Score

The models were trained on 1,000 airline flight records, and the best-performing model was selected based on accuracy and generalization performance for deployment via API.

Model Evaluation Metrics and Results

Model Evaluation Metrics and Results

The trained models were evaluated using standard regression metrics to measure performance and accuracy in predicting meal demand and food waste.

Evaluation Metrics:

R^2 Score (Goodness of fit)
Mean Squared Error (MSE)
Mean Absolute Error (MAE)

Results Summary:
Multiple models were compared, and ensemble-based models (Random Forest, XGBoost, Gradient Boosting) showed the best performance in terms of accuracy and stability. The final selected model achieved the highest R^2 score and lowest prediction error, making it suitable for deployment.

Overall, the results confirmed that machine learning can effectively predict airline meal requirements and reduce food waste with high reliability.

Deployment Approach (e.g., API deployment)

Deployment Approach (Hugging Face API)

The trained machine learning model was deployed using the Hugging Face platform to provide real-time predictions through an API.

After finalizing the best-performing model, it was serialized and uploaded to a Hugging Face inference endpoint, enabling seamless integration with external applications.

The deployment architecture allows users to send flight-related inputs (such as passenger count, flight duration, and aircraft type) via API requests, and receive instant predictions for:

Meal demand
Recommended food quantities
Estimated food waste percentage

This approach ensured scalability, fast response time, and easy integration with the conversational chat interface built on top of the system.

Description of the Application Layer (e.g., chatbot, interface)

Description of the Application Layer (Chatbot Interface)

The application layer was developed as an intelligent conversational interface that allows users to interact directly with the machine learning model in a simple and user-friendly way.

A chatbot-based system was built to collect flight details such as passenger count, flight duration, aircraft type, and departure time. These inputs are then sent to the deployed Hugging Face API in real time.

The chatbot processes the response and returns:

Predicted meal requirements
Suggested food quantities
Estimated food waste percentage

This layer acts as a bridge between the user and the AI model, removing the need for technical knowledge and enabling fast decision-making for airline catering operations.

Testing and Iteration Processes Used for Improvement

The system was tested iteratively to ensure accuracy, stability, and reliable real-time predictions.

Testing Approach:

Unit testing for individual components (data processing, model prediction, API calls).
End-to-end testing of the full pipeline from user input to prediction output.
Validation using unseen test data to measure real-world performance.

Iteration Process:

Model performance was analyzed after each training cycle.
Hyperparameters were tuned to improve accuracy and reduce error.
Feature selection was refined to remove less impactful variables.
API responses were optimized for speed and reliability.
Chatbot prompts and response formatting were improved based on testing feedback.

This iterative approach ensured continuous improvement of both model performance and user experience, resulting in a stable and efficient AI-driven system for airline meal prediction and food waste estimation.

Methodologies Followed (e.g., Agile, MLOps)

The project was developed using a combination of Agile methodology and MLOps practices to ensure structured development, fast iteration, and reliable deployment.

Agile Methodology:
The project was divided into short iterative sprints, each focusing on a specific stage of development:

Sprint 1: Problem definition and requirements gathering
Sprint 2: Data collection and preprocessing
Sprint 3: Model training and experimentation
Sprint 4: Model evaluation and selection
Sprint 5: API deployment and integration
Sprint 6: Chatbot development and user interface integration
Sprint 7: Testing, feedback, and optimization

This approach allowed continuous improvement and flexible adjustments throughout the development cycle.

MLOps Practices:
MLOps principles were applied to bridge the gap between model development and production deployment:

Version control of datasets and models
Deployment of the trained model via Hugging Face API
Integration of the model into a real-time application layer (chatbot)
Continuous testing and performance monitoring
Iterative improvements based on system feedback

This combination ensured a scalable, maintainable, and production-ready AI system for airline meal prediction and food waste optimization.

Continuous Improvement or Monitoring Strategies Implemented

A continuous improvement approach was applied to ensure the system remains accurate, reliable, and adaptable over time.

Model performance monitoring: Prediction outputs were regularly reviewed to detect any drop in accuracy or inconsistencies.
Error analysis: Mispredictions were analyzed to identify weak patterns in the data and improve feature selection.
Feedback loop: Insights from testing and real usage were used to refine the model and chatbot responses.
Retraining strategy: The model can be retrained periodically using updated flight data to improve generalization.
API monitoring: Hugging Face API performance (latency and response stability) was observed to ensure smooth integration.
Incremental improvements: Small iterative updates were applied to preprocessing steps, features, and model parameters based on observed results.

This strategy ensures the system stays effective over time and continuously improves in predicting meal demand and reducing food waste.

Start Point Workflow	**On Submission** Form	Jun 20, 2026 12:38 AM